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Group Number& Subject	1623
Date	8 may 2026

1.Project is for Simulation only.

2. Check the Excel file and fill the name of the project. See the project is not repeated, and it should not be taken from the Practical Manual, if the project is repeated change different Simulator.
(like Proetus, Tinker cad or Arduino Simulator) .

Attach is the Excel Sheet. [1622.xlsx](#)

3.Attach is the Project Report and complete it and submit on time.
(Last date of Submission of project is on 10/05/2026.

4.Maximum Marks are 5.

5.Attach the Screenrecording, when Implementing the project using build in recording App or Clipchamp, Cap cut or any other App.
(Without any audio or text,just to know you are only doing it.).

6.If the recording is long ,create a link and share through the google drive.

7.If any other issue contact personally in the Lab.

8.Submit the Project Report in the word format only.

MINI PROJECT REPORT

Embedded Systems Lab

Title:

Smart Hospital Room System

1. Introduction

Technology has become an important part of modern healthcare systems. Smart hospital rooms help improve patient safety, reduce manual monitoring, and assist medical staff in responding quickly to different situations.

This project presents a Smart Hospital Room System using Arduino UNO and Proteus simulation. The system monitors the IV fluid level, allows patients to request a nurse using a push button, and automatically controls the cooling system according to room temperature.

The project demonstrates how embedded systems can improve healthcare environments through automation and real-time monitoring.

Video:

<https://drive.google.com/drive/folders/1t81VjbT7i9P0WCY28NAjHbLGmFqj0QMp?usp=sharing>

2. Objectives

The objectives of this project are:

- .To monitor the IV fluid level using an ultrasonic sensor
- .To alert medical staff when the IV level becomes low
- .To provide a nurse call system using a push button
- .To monitor room temperature using an LM35 temperature sensor
- .To automatically activate the cooling system when the temperature increases
- .To display system status messages on an LCD screen

3. Components Used

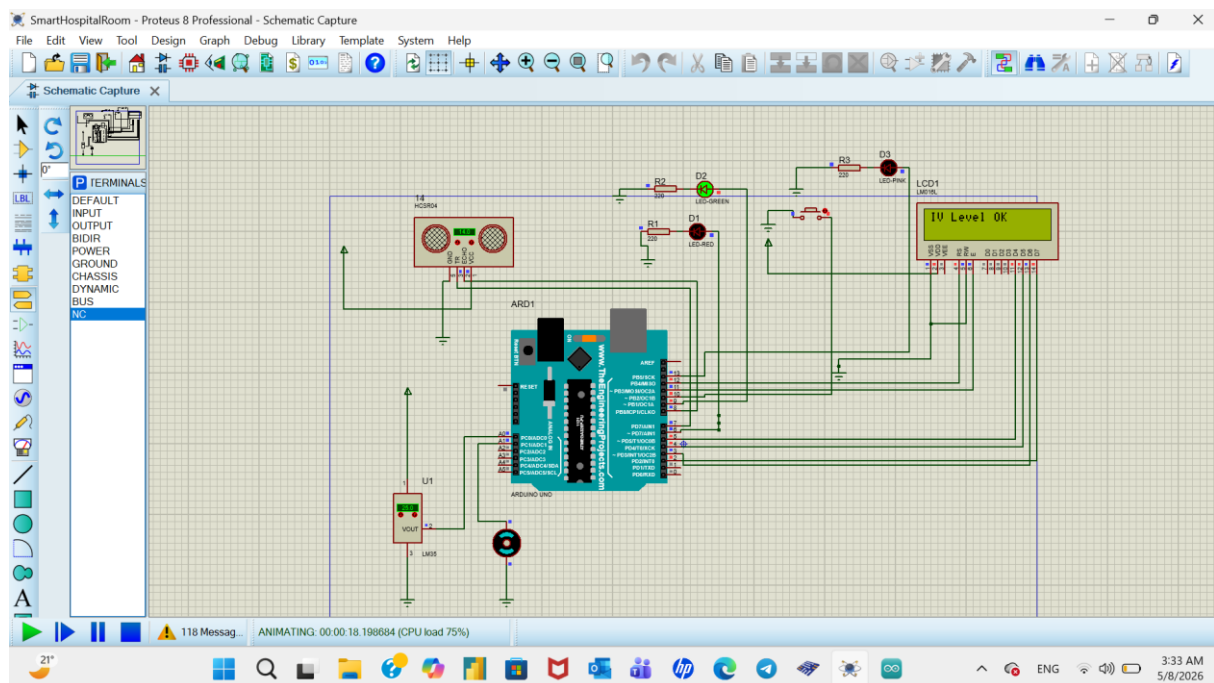
Components Used

Component	Specification	Purpose
Arduino UNO	ATmega328P	Main controller of the system
HC-SR04 Ultrasonic Sensor	Distance Sensor	Monitor IV fluid level
LM35 Temperature Sensor	Analog Temperature Sensor	Measure room temperature
LCD Display	16x2 LCD	Display system messages
Push Button	Digital Input	Nurse call request
Red LED	5V LED	Indicate low IV level
Green LED	5V LED	Indicate normal IV level
Pink LED	5V LED	Nurse assistance indication
DC Motor	Small DC Motor	Represent cooling system
Resistors	220 Ohm	Protect LEDs and circuit
Jumper Wires	Connection Wires	Circuit connections
Proteus Software	Simulation Tool	Circuit simulation

4. Block Diagram

:The Smart Hospital Room System consists of

- .Arduino UNO as the main controller
- .HC-SR04 sensor connected to monitor IV level
- .LM35 sensor connected to monitor room temperature
- .Push button connected for nurse assistance requests
- .LCD connected to display system status
- .LEDs used for IV status indication
- .DC motor used as a cooling system



5. Methodology

The system works as follows:

- The ultrasonic sensor continuously measures the IV fluid level.
- If the measured distance becomes greater than the specified limit, the system detects a low IV level.
- A red LED turns ON and a warning message appears on the LCD.
- If the IV level is normal, a green LED turns ON.
- The patient can press the nurse button to request assistance.

When the button is pressed, the LCD displays “Nurse Needed” and the nurse LED blinks.

The LM35 sensor continuously measures room temperature.

If the temperature exceeds 30°C, the cooling system automatically turns ON.

When the temperature returns to normal, the cooling system turns OFF.

6. Circuit Diagram (Description)

.Insert Proteus circuit screenshot here

.The circuit was designed and simulated using Proteus software

.The HC-SR04 ultrasonic sensor is connected to digital pins 7 and 8

.The LCD display is connected using digital pins 12, 11, 5, 4, 3, and 2

.The nurse button is connected to pin 10 using INPUT_PULLUP mode

.The red LED is connected to pin 6

.The green LED is connected to pin 9

.The nurse LED is connected to pin 13

.The LM35 temperature sensor output is connected to analog pin A0

.The cooling system (DC motor) is connected to analog pin A1

7. Arduino Code (Sample)

```
lab5.ino
1  #include<LiquidCrystal.h>
2  LiquidCrystal lcd(12,11,5,4,3,2);
3
4  int TrigPin=7;
5  int EchoPin=8;
6  int RedLed=6;
7  int GreenLed=9;
8  int NurseLed=13;
9  int NurseButton=10;
10 int TempPin=A0;
11 int CoolingPin=A1;
12
13 long duration;
14 int distance;
15
16
17 void setup() {
18   pinMode(TrigPin,OUTPUT);
19   pinMode(EchoPin,INPUT);
20   pinMode(RedLed,OUTPUT);
21   pinMode(GreenLed,OUTPUT);
22   pinMode(NurseLed,OUTPUT);
23   pinMode(CoolingPin,OUTPUT);
24   pinMode(TempPin,INPUT);
25   pinMode(NurseButton,INPUT_PULLUP);
```

lab5.ino

```
25 pinMode(NurseButton, INPUT_PULLUP);
26
27
28 lcd.begin(16,2);
29 }
30
31 void loop() {
32
33     // IV Monitoring
34
35     digitalWrite(TrigPin, LOW);
36     delayMicroseconds(2);
37
38     digitalWrite(TrigPin, HIGH);
39     delayMicroseconds(10);
40
41     digitalWrite(TrigPin, LOW);
42
43     duration = pulseIn(EchoPin, HIGH);
44
45     distance = duration * 0.034 / 2;
46
47     lcd.clear();
48
49     if(distance > 15){
```

```
        if(distance > 15){
            lcd.print("IV Level LOW");
            digitalWrite(RedLed,HIGH);
            digitalWrite(GreenLed,LOW);
        }
        else{
            lcd.print("IV Level OK");
            digitalWrite(RedLed,LOW);
            digitalWrite(GreenLed,HIGH);
        }

        // Nurse Button

        int nurseState = digitalRead(NurseButton);

        if(nurseState == LOW){
```

Screen
Auto

8. Results

Insert simulation screenshots here.

The Smart Hospital Room System was successfully simulated using Proteus.

The following results were achieved:

The system correctly monitored the IV fluid level.

The red LED turned ON when the IV level became low.

The green LED turned ON when the IV level was normal.

The nurse button successfully displayed “Nurse Needed” on the LCD.

The nurse LED blinked when assistance was requested.

The LM35 sensor successfully monitored room temperature.

The cooling system automatically activated when the temperature exceeded 30°C.

9. Applications

This project can be used in:

Hospitals

Smart healthcare systems

Patient monitoring systems

Home healthcare environments

Elderly care systems

10. Conclusion

The Smart Hospital Room System demonstrated the importance of embedded systems in healthcare applications. The project successfully integrated monitoring, automation, and alert systems using Arduino UNO and Proteus simulation.

The system improved patient safety and provided automatic responses for important situations such as low IV levels and high room temperatures.

11. Future Enhancements

Future improvements may include:

- Adding WiFi or Bluetooth communication.
- Sending notifications directly to nurses.
- Adding heart rate and oxygen monitoring sensors.
- Developing a mobile application for monitoring.
- Using IoT technology for remote healthcare systems.